Fiscal Year:	FY 2005	**Task Last Updated:**	FY 11/02/2005
PI Name:	Lupton, Joanne R. Ph.D.		
Project Title:	Nutritional Countermeasures to Radiation-Enhanced Colon Cancer		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Nutrition, Physical Fitness, and Rehabilitation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77843-2253	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	10/01/2004	**End Date:**	09/30/2008
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Braby, Leslie (Texas A&M University) Chapkin, Robert (Texas A&M University) Ford, John (Texas A&M University) Turner, Nancy (Texas A&M University) Carroll, Raymond (Texas A&M University)		
Grant/Contract No.:	NCC 9-58-NPFR00402		
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Task Description:	1. Original Aims. The primary goal of this project is to develop a set of colon cell gene expression profiles that accurately characterize the radiation-enhancing effect on colon tumor development and the chemoprotective effect of dietary countermeasures in this process. This goal is addressed by specific aim #1: From the global transcriptional profile (analyzed by microarray technology) in samples isolated from colonic mucosa, determine which sets of gene expression profiles best characterize the initiation stage, the promotion stage (aberrant crypt development) and the final tumor stage of colon cancer development using a 2 x 5 x 4 experimental design (+ or – radiation; five time points and 4 diets). The secondary goal, addressed by specific aim #2, is to compare gene expression profiles developed from exfoliated cells to those from colonic mucosal cells to determine how well they reflect in vivo events. This information is important for our short-term future goal of using fecal material from humans as a noninvasive method to monitor changes in gene expression patterns over time. Our long-term future goal is to have sets of gene expression profiles that can be used to monitor astronauts before, during, and after space flight to detect exposure and response to a radiation event so that appropriate intervention strategies (e.g., dietary countermeasures) can be implemented as needed. Data generated from this research program will support the NASA/NSBRI goals of 1) gaining greater understanding of the mechanisms involved in the synergy of radiation and other environmental insults on carcinogenesis, 2) developing a system to monitor the need for countermeasure administration during or after flight, and 3) development of countermeasures that can be applied before, during and after flight. 2. Key Findings. We have optimized the RNA isolation procedures so that we routinely obtain sufficient quantities of RNA that is of very high quality so that microarray analyses are possible. 3. Impact of Findings on Project Goals. The work to date has us well on track to achieve the goals described in specific aim #1 of this project. 4. Proposed Research Plan for the Coming Year. The isolated mucosal RNA will start being used for microarray analyses. To prepare for this we will need to perform the appropriate reverse transcription reactions and label the RNA in preparation for array hybridizations. Results from the microarray analyses will be statistically analyzed and key genes identified so that PCR assays may be initiated in order to confirm the microarray results.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results generated by this project will extend to Earth-based medical care. Development of a set of genes that are indicative of colon cancer development, and that can be monitored through a noninvasive procedure would greatly increase screening success and reduce the current high rates of morbidity and mortality associated with this disease.
Task Progress:	In the first year we have tested a variety of modifications to the mucosal RNA isolation procedures in order to optimize the quality of the RNA to be used in the reverse transcription reactions. After the final protocol was developed we have successfully isolated RNA from all the mucosal samples to be used in this project. Both the quality and quantity of RNA in the isolates have been determined. The first steps in the microarray analyses will be initiated in August, 2005.
Bibliography Type:	Description: (Last Updated: 08/22/2020)
Articles in Peer-reviewed Journals	Hong MY, Bancroft LK, Turner ND, Davidson LA, Murphy ME, Carroll RJ, Chapkin RS, Lupton JR. "Fish oil decreases oxidative DNA damage by enhancing apoptosis in rat colon." Nutr Cancer. 2005;52(2):166-75. PMID: 16201848 , Jan-2005
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Presentation	Hong, M. Y., N. D. Turner, M. E. Murphy, R. J. Carroll, R. S. Chapkin, and J. R. Lupton "Dietary fish oil may decrease colonic cell proliferation by upregulating p27Kip1 level in actively proliferating cells" N/A Apr-2005
Presentation	Hong, M. Y., N. D. Turner, M. E. Murphy, R. J. Carroll, R. S. Chapkin, and J. R. Lupton "Dietary fish oil reduces N7-deoxymethylguanine adducts by enhancing apoptosis at the initiation stage of carcinogenesis in the small (SI) and large intestine (LI)" N/A Apr-2005
Presentation	Hong, M. Y., N. D. Turner, R. J. Carroll, R. S. Chapkin, and J. R. Lupton "Higher DNA adduct levels and lower apoptosis may explain the higher cancer susceptibility in the distal colon compared to the rest of the gastrointestinal tract" N/A Apr-2005

Presentation	Hong, M. Y., N. D. Turner, S. S. Taddeo, M. E. Murphy, R. J. Carroll, R. S. Chapkin, and J. R. Lupton "Influence of dietary lipid and butyrate on the coordination of proliferation differentiation and apoptosis in the same cell during the initiation stage of colon carcinogenesis" N/A Apr-2004
Presentation	Lupton, J. R., L. M. Sanders, J. C. Mann, N. Popovic, A. A. Glagolenko, L. A. Davidson, L. A. Braby, J. R. Ford, R. J. Carroll, N. Wang, R. S. Chapkin, and N. D. Turner "A combination of omega-3 fatty acids and a butyrate-producing fiber mitigates colon cancer development" N/A May-2005
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Presentation	Turner, N. D., C. A. Warren, K. J. Paulhill, L. M. Sanders, M. Y. Hong, K. L. Covert, L. A. Davidson, R. S. Chapkin, and J. R. Lupton "Fermentation products in colon health: Mediators of cell kinetics and gene expression" N/A Apr-2005
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